AEL Technical Blueprint

1 How the tool works

The tool is a Python program that we use through the terminal. It follows a fixed order of steps and uses a local language model through Ollama. We provide the Dutch texts ourselves. The program does not search the internet or automatically download articles from URLs.

2 The main parts

Part	Main job
Translation agent	Checks for personal details, translates Dutch to English, simplifies the language and reviews the translation.
Quality checks	Compare the translation with the Dutch input and check numbers, placeholders and meaning.
File saver	Saves the English TXT together with a .source.json file containing source details and checking information.
Handbook agent	Checks the chosen translations and creates a handbook with evidence, references and suggested next steps.
Wiki archive	Keeps the received Dutch input and final English translation together in a local browser view.
Git publisher	Uploads the selected handbook PDF to the handbooks folder on GitHub.

3 What passes between the parts

The translation agent receives a Dutch text and produces an English text with source details. The saved translation and metadata file must stay together. The handbook agent reads the selected saved translations. It produces a PDF, a Markdown version and a quality report.

The archive is updated separately after translation. It is not an input to the handbook. If archiving fails, the tool reports a warning and the main workflow can continue.

4 How the checks work

If the translation checks find a problem, the program tries to repair it once. If the problem remains, it stops. This happened with the longer CBS article, which was not used in our final demonstration.

The handbook checks that its supporting quotations exist in the supplied English sources. A model also checks the meaning of proposed statements. If that meaning check rejects the statements, the tool can replace them with exact passages from the translations.

This replacement is recorded in the quality report. A final PASS can therefore mean that exact passages were accepted after the meaning check failed. It is not proof that all source information is true. Errors already present in a translation can still appear in the handbook.

5 Running and publishing

The setup needs Python, Ollama with the selected model, the Markdown package, wkhtmltopdf for PDF creation and Git. The default model is qwen2.5:3b. From the AI_Agent folder, we start the workflow with python dutch_translation_agent.py.

The user chooses whether to save, create a handbook and publish. Before publication, the user should inspect the PDF and quality report. The publisher uses a separate temporary copy of the GitHub repository and uploads only the selected PDF. It does not publish the local wiki or all project changes.

GitHub login and write access are required. If publication fails, the generated files remain available for retry. Our demonstration video shows a successful upload of the new PDF. This is repository publication, not automatic deployment of a GitHub Pages website.

6 Limits of the final version

Private interviews and company records are outside this workflow. The archive contains the input before anonymisation. The model is local by default, but settings can change that. We have not fully pinned all software versions or proven identical results on every laptop.